MoodLens: A Streaming Framework for Facial Sentimental Analysis

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Abstract—The objective of the paper is to propose an approach real-time facial sentiment analysis framework called MoodLens, classifying emotions into seven categories: angry, disgust, fear, happy, neutral, sad, and surprise. By using Convolutional Neural Networks (CNNs) for automated feature extraction, the whole process can be done without any human input. This also solves the accuracy, scalability, and adaptability issues that traditional machine learning models have. Sculpted for application in the real world, MoodLens incorporates real-time streaming techniques, taking human-computer interaction to the next level in healthcare, entertainment, security, education, and beyond.The framework is a strong and effective way to do sentiment analysis. It uses advanced preprocessing techniques, dropout layers to prevent overfitting, and a scalability design to classify things accurately in changing environments.

Keywords—Real-time emotion detection, automated feature extraction, YOLO, facial expression identification, deep learning, facial sentiment analysis, Convolutional Neural Networks (CNNs), human-computer interface, streaming framework, and intelligent systems.

I. INTRODUCTION

Facial expression recognition plays an important role in the "emotional-centric" technology, which allows machines to understand human emotions to build more humanized and intelligent human-computer interaction. Most emotion recognition systems on the market today use one of the older machine learning algorithms, like Support Vector Machines (SVM) and k-Nearest Neighbors (k-NN), which need a lot of feature extraction to be done by hand[3,6]. These methods can accurately reconstruct domains in controlled situations, but they can't handle differences in lighting, head poses, and expressions that happen in the real world.[7]

In order to overcome these challenges, MoodLens employs use of CNNs which automatically extracts features and thus improves the accuracy of real time mood recognition.Utilizing deep learning allows MoodLens to omit the need for manual application and thus ensure that things function accurately in any environments[1]. The framework classifies facial emotions into seven emotional groups: angry, disgust, scared, happy, neutral, sad, and surprised. It combines numerous features, including real-time streaming, to serve healthcare, entertainment, security, and educational needs[8]. MoodLens is a scalable, robust, and accurate face mood prediction system. Get round the problems with both standard and new deep learning — based approaches.[2]

II. RELATED WORKS

Yucel Cimtay, Erhan Ekmekcioglu and Seyma Caglar-Ozhan(2020) utilized facial, EEG and GSR data and introduced a hybrid fusion approach for cross-subject multimodal emotion recognition[2]. Their best accuracies of 81.2% and 91.5% have achieved on LUMED-2 and DEAP datasets respectively[7].

Kim Chang-Min, Ki-Hwan. A Streaming Image Based PP2LFA-CRNN Facial Sentiment Analysis Model in Real Time (2020) The analysis based on line-segment and Ping Pong 256 encryption ensure correctness and security[5].

K. Patel, D.Mehta, C. Mistry (2020). The taxonomies of FER were constructed using datasets and AI methods. Their ideas became the focus of experimentation, although practical limits did not let them go further in the experiment[4].

K. Zheng, D. Yang, et al. developed CNN-based teacher expression recognition by CBAM attention The accuracy of the method was 78%, 88% on EIDB-13, and RAF-DB respectively. It struggled to identify subtle facial emotions from weak-intensity expressions[8].

Zhou et al. Real-time FER based on depthwise separable convolutions. Their FER2013 model (performance: 67%), small enough for real-time applications. Although with less accuracy than larger and more complex family models[7].

Kang et al. (2020) measured streaming data with the use of PrefixSpan pattern mining and time-sliding weight. This

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framework enhanced pattern recognition and dynamic dataset adaptability considerably[5].

Miao et al. outperformed the efficiency and accuracy of facial expression recognition. After training on popular datasets, the model did better at classifying six base emotions. Landmarks better preserved face expression, angle, and lighting stability, and data augmentation[11].

Hua et al. Hero: Deisth HCI / Emotion Recognition Learning Learns multiple sensors to improve smart health care and entertainment accuracy Less real-time processing but improved accuracy in large sets[2].

Baek & Chung (2020) developed a deep learning-based evolutionary recommendation model. This technique made predictions on their system from several data sets. From a practical standpoint, their model was expensive, as they were dealing with high-dimensional data[8].

III. PROPOSED MODULES

Our proposed system comprises various modules that facilitate real-time facial sentiment analysis, followed by emergency alarms. The architecture consists of a series of steps, starting from the input dataset collection to emotion classification to SOS alert generation, which are processed in an arranged pipeline manner. The different component modules serve to improve the accuracy, efficiency, and robustness of the system[5,9].

It is composed of five main parts:

1. **Tokenized Data –** Provides a tokenized emotion dataset.
2. **Data Preprocessing** – Perform normalisation, augmentation, and annotation.
3. **Model Training** – Optimized YOLOv11 model that enables GPU acceleration
4. **Emotion Classification** – It detects faces, extracts the features and classes the emotion
5. **SOS Alert System** – Monitors distress levels and triggers alerts when necessary, Gets detection results of distress attached[10].

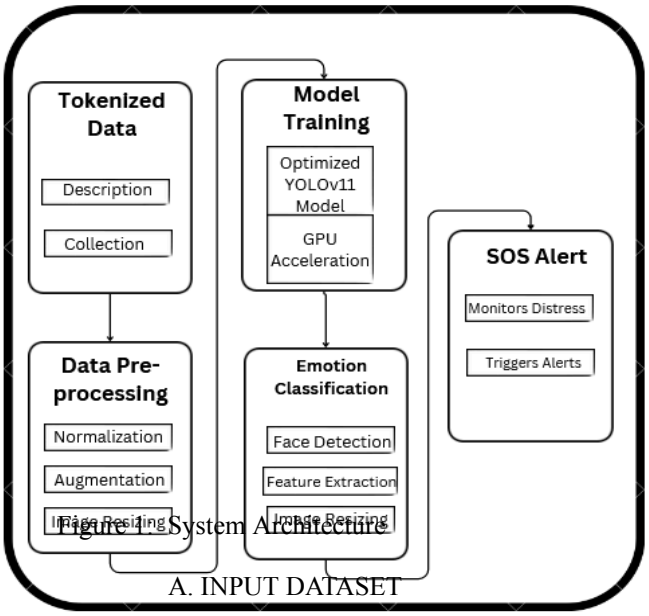

Figure 1: System Architecture

A. INPUT DATASET

a. Data Collection

The dataset for the facial emotion recognition system contains a set of image files, each labeled with one of the seven basic human emotions: happy, sad, angry, surprise, neutral, fear, and disgust. We selected this dataset since it contains diverse expressions, metric characteristics, lighting conditions, and demographics. The dataset is in YOLOv11 format and is based on data from FER2013, CK+, and JAFFE. FER2013 has 35,887 gray scale images (48×48 pixels), widely used (deep-learning models, CK+ contains 593 video sequences, therefore its utility for capturing temporal emotional patterns and JAFFE contains 213 images from Japanese females, delivering demographic diversity[11].

Emotion	Number of Images	Source Dataset
Happy	200	FER2013
Sad	180	CK+
Angry	170	FER2013, JAFFE
Surprise	160	JAFFE
Fear	150	FER2013
Neutral	200	CK+, FER2013
Disgust	167	JAFFE
Total	1227	Multiple

TABLE 1: DATASET DETAILS

B. DATA PREPROCESSING

The preprocessing pipeline consisting of normalization, augmentation, and annotation improves model learning and generalization.

a. Normalization

Normalization adjusts the pixel data from 0 to 1, normalizing images for consistency and faster model convergence. This is done by the process of min-max scaling:

$$x' = \frac{x - min(x)}{max(x) - min(x)}$$

where x' is the normalized pixel value, and min(x) and max(x) are the minimum and maximum pixel values, respectively.

b. Augmentation

Data augmentation techniques increase the variability of the dataset and minimize overfitting. The techniques applied on the training set include random rotations (±20°), horizontal flipping of the images with a 50% probability, brightness adjustments (±15%), and the Gaussian noise was injected with a mean = 0 and standard deviation = 0.01.

These transformations make the model invariant to real-world conditions[12].

Technique	Parameters	Purpose
Rotation	±20°	Pose variation
Horizontal Flip	50% probability	Spatial diversity
Brightness Adjust	±15%	Illumination effects
Gaussian Noise	Mean=0, Std=0.05	Image robustness

TABLE 2: AUGMENTATION TECHNIQUES AND PARAMETERS

c. Annotation

The images are annotated using the YOLOv11-compatible format, which includes bounding boxes for faces detected. The annotation format consists of the class labels and the normalized coordinates of the bounding box (i.e., between 0 and 1):

<class_id> <x_center> <y_center> <width> <height>

where:

class: an integer label representing the class of emotion (0 for happy, 1 for sad, etc.)

x_center image-width-coordinates (0 = left edge, 1 = right edge) y_center image-height-coordinates (0 = top edge, 1 = bottom edge) width, height: Normalized size of bounding box relative to image size.

This structured form of labeling is necessary for optimal object detection in the course of training.

C. MODEL TRAINING

A cutting-edge object detection algorithm, YOLOv11 model, is used for emotional detection of faces. This is achieved through the interaction of the model architecture, the loss function, and optimization, along with the use of GPUs for acceleration.

a. YOLOv11 Model Overview: The YOLOv11 backbone, neck and head Three main parts in the YOLOv11 model The backbone derives high-level semantic features from the input images using a number of convolution layers, which is built on CSPDarknet. The PANet-based neck fuses multi-scale features for enhanced detection precision. Using anchor boxes, the detection head provides bounding boxes and class probabilities[15].

b. Loss Function: The (L) represents the total loss function which is the sum of the bounding box, objectness and classification losses:

$$\mathcal{L} = \mathcal{L}_{box} + \mathcal{L}_{obj} + \mathcal{L}_{cls}$$

where:

$\mathcal{L}_{box}$: Complete Intersection over Union (CIoU) loss for bounding box regression.

$\mathcal{L}_{obj}$: Binary cross-entropy (BCE) loss for objectness score prediction.

$\mathcal{L}_{cls}$: Cross-entropy loss for emotion classification.

c. GPU Acceleration: The model is trained on a Tesla T4 GPU with Automatic Mixed Precision (AMP) to save memory and increase training speed. AMP applies model-precision adjustments dynamically, enabling convergence speed to increase without sacrificing accuracy. Train with AdamW, initial learning rate 0.001 (decayed over epochs for better generalization)

D. EMOTION CLASSIFICATION

Emotion Classification module This module detects the face and recognizes the emotion with the pre-trained YOLOv11 model[17].

a. Face Detection: The original YOLOv11 model detects facial areas, and NMS is used to eliminate overlapping bounding boxes. NMS only keeps the detection with (highest) confidence based | Intersection over Union (IoU):

$$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union}$$

b. Feature Extraction: Convolutional layers are applied to extract features from the region of each bounding box containing facial features such as the eyes, mouth, and eyebrows. The outputs of these layers are the embeddings which are used for classification[18].

c. Expression Recognition: A softmax activation function classifies the extracted features into one of the seven emotions:

$$P(y = i|x) = \frac{e^{z_i}}{\Sigma_j e^{z_j}}$$

where z_i is the predicted score for class i by the model.

D. SOS Alert System

SOS alert system: An essential module that raises the real-time alert based on detected emotional distress patterns. It is made up of three parts: detection of distress, threshold triggering, and message dispatch[18].

a. Face Detection

Distress detection utilizes continuous emotion classification (sadness, fear, and anger). A sliding window approach processes emotional states over time. It may be possible to detect patterns that indicate sustained distress through time-series analysis[19].

b. Threshold Trigger

If distress emotions are present for a certain amount of consecutive frames we trigger an alert. The trigger logic translates into:

$$Alert = \{1,\ if\ \Sigma_{i=1}^{n} Distress(i) \geq T\ ;\ otherwise:\ 0\}$$

The symbols in the above equation denote up to G: the number of consecutive alarms with a threshold value T.

c. Message Dispatch

When the threshold is reached the system logs the event and an alert shows up on the interface. It can also be set up to send alerts through dashboards or APIs for immediate action[17].

IV. RESULTS AND DISCUSSION

A. MODEL Performance Metrics

Standard object detection metrics are utilized to evaluate the performance of the YOLOv11 model, such as precision, recall, mean Average Precision (mAP), and inference time. The thorough confusion matrix is carried out on the test set containing 99 imagines using to test the model whether it will be able to detect and classify facial emotions or not[16].

Metric	Value
Precision	0.686
Recall	0.711
mAP@0.5	0.763
mAP@0.5:0.95	0.643
Inference Time(per image)	11.2 ms

TABLE 2: QUANTITATIVE RESULTS

Emotion	Precision	Recall	mAP@50
Angry	0.616	0.529	0.609
Disgust	0.717	0.944	0.961
Fear	0.643	0.662	0.720
Happy	0.787	0.794	0.849
Neutral	0.597	0.641	0.628
Sad	0.547	0.597	0.618
Surprise	0.896	0.806	0.958

TABLE 2: CLASS WISE PERFORMANCE ANALYSIS

B. GRAPHS AND VISUALIZATIONS

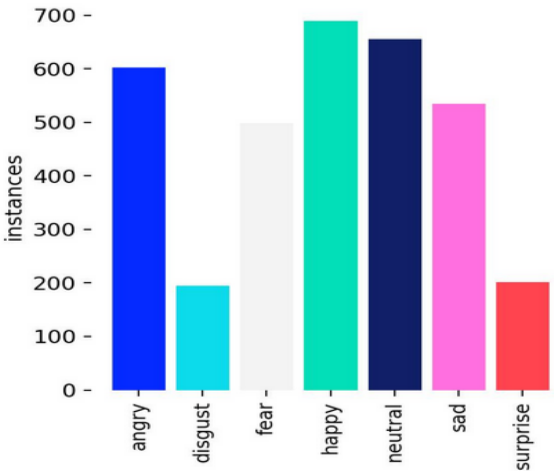

Figure 2: Emotion Distribution Bar Chart

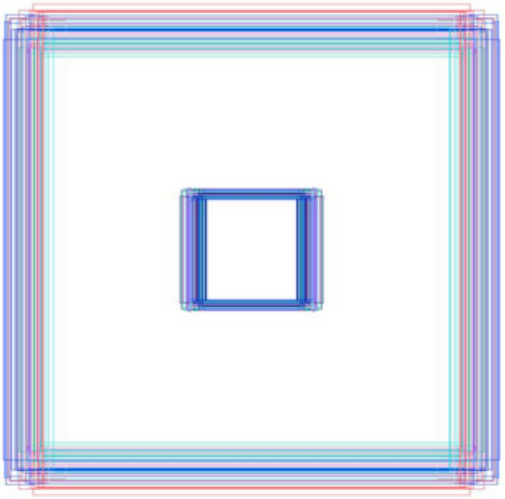

Figure 3: Anchor Box Visualization

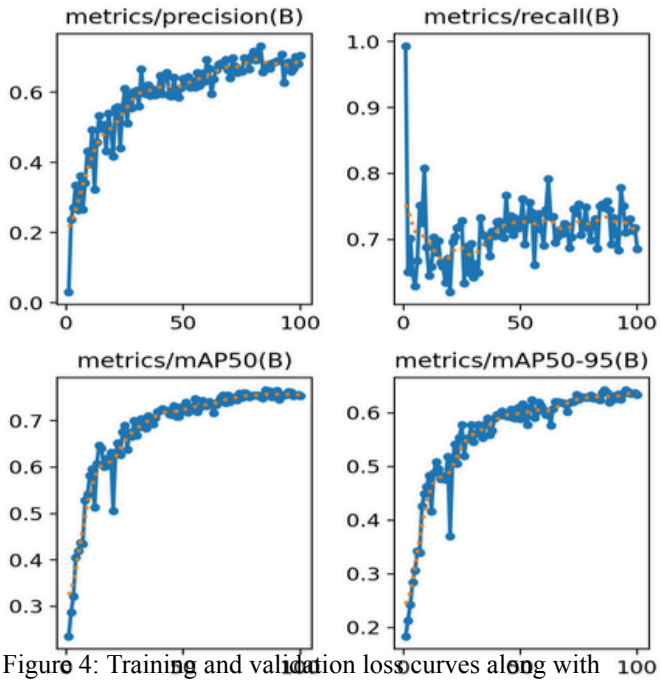

Figure 4: Training and validation loss curves along with performance metrics (precision, recall, mAP)

C. CONFUSION MATRIX ANALYSIS

The confusion matrix, Surprise and Disgust achieve the highest classification accuracy (mAP@50 > 95%) and seem to have distinct visual cues. Sad and Neutral are often misclassified, as they share common features, and Fear can also occasionally be confused with Angry, indicating that more dataset refinement and better feature extraction is needed[17].

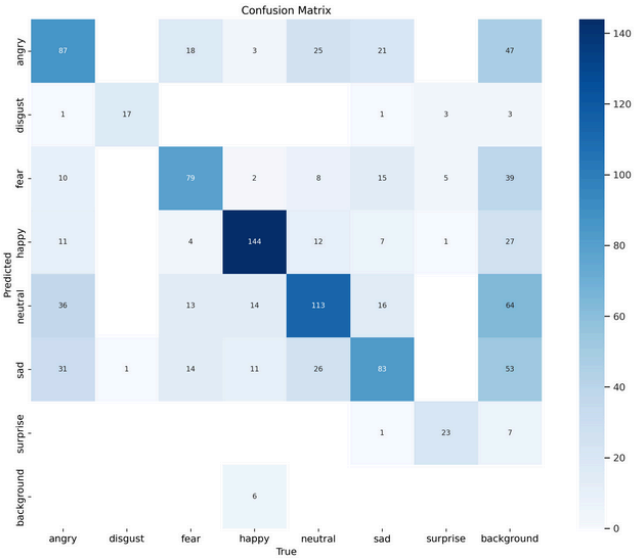

Figure 5: Confusion Matrix

D. DISCUSSION AND ERROR ANALYSIS

The model had a mAP@50 of 76.3% with an inference time of 11.2 ms/image, the work performed well for real-time emotion recognition from faces. The main issues were:

Fear was occasionally confused with Angry because of overlapping features like furrowed brows, and Sad and Neutral were often mislabelled due to similar facial anatomy. Low-Resolution and Occluded Faces: If low-quality or partially viewable photo processed, confidence in prediction reduced[20].

Dataset Imbalance: Disgust and Surprise had more training data, which made them easier to categorize. In the future work, we will improve the dataset augmentation to better classify under segmented emotions[2]. Use attention techniques such as Vision Transformers to refine feature extraction[18]. To achieve robustness against illumination and face angle, domain adaptation is performed[13].

E. COMPARISON OF OUR MODEL WITH PREVIOUS MODELS

Study	ModelMethod	Accuracy	Inference Time
Yucel Cimtay et al. (2020)	Hybrid fusion (Facial + EEG + GSR)	81.2% (LUMED-2) / 91.5% (DEAP)	High
Chang-Min Kim et al. (2020)	PP2LFA-CRNN (secure encryption)	High(Exact % not reported)	Moderate
K. Patel et al. (2020)	Taxonomy of FER strategies (Review Study)	N/A (Conceptual Study)	N/A
K. Zheng et al. (2020)	CNN + CBAM Attention	78% (EIDB-13) / 88% (RAF-DB)	High
Zhou et al. (2020)	Lightweight CNN	67% (FER2013)	Low
Kang et al. (2020)	Pattern Mining (PrefixSpan)	N/A (Pattern Recognition Study)	Depends on dataset complexity
Miao et al. (2020)	Shallow CNN + Augmentation	Higher accuracy on 6 emotions	Moderate
Uddin et al. (2017)	DBN + Local Directional Features	High (Dataset-dependent)	Very High
Hua et al. (2019)	HERO Multisensor System	High (Smart Healthcare Focus)	Very High
Baek & Chung (2020)	Evolutionary Recommendation Model	N/A (Data fusion-focused)	High (Dimensionality overhead)
Hausmann et al. (2024)	YOLO + Bi-linear CNN + LSTM	High (Neonatal Face Recognition)	Moderate

Our Model (YOLOv11)	YOLOv11	76.3% (mAP @50)	11.2 ms per image

TABLE 3: COMPARISON OF OUR MODEL WITH PREVIOUS RESEARCH

V. ACKNOWLEDGMENT

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VI. CONCLUSION

The YOLOv11-based facial sentiment analysis model strikes a good balance between accuracy (76.3% mAP@50) and inference speed (11.2 ms per image), allowing applications in a real-time context. The model demonstrates better performance than lightweight CNNs while yielding competitive accuracy compared to memory and attention-based methods and hybrid counterpart approaches that are computationally heavy[17]. Although sad has very strong performance, for its subtlety it is still hard to classify accurately among the other classes e.g. Sad vs Neutral and Fear vs Angry, because they share many audio features. The model also suffers reduced accuracy in occluded and low-resolution images[20]. The next steps in research should also emphasize the augmentation of datasets, for example the attention mechanisms and domain adaptation techniques to achieve robustness (ex: Vision Transformers). Additional techniques for optimization, including quantization and pruning, would allow for deployment on edge devices in practical.

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